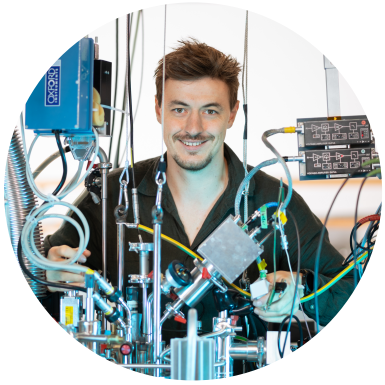
Oliver Aschenbrenner

(formerly Irtenkauf)



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Education

Doctorate in Physics (ongoing)

Thesis title: Superconducting Atomic Contacts under Microwave Irradiation

2021 – now | Konstanz, Germany

M.Sc. in Physics

Thesis title: Broadband Ferromagnetic Resonance Measurements on Thin Cobalt Films at Low Temperatures

2017 – 2021 | Konstanz, Germany

B.Sc. in Physics

Thesis title: Influence of Reciprocity of One-Dimensional Anderson Localization of Light

2013 – 2017 | Konstanz, Germany

Experience

Erasmus at Umeå University

Advanced Project: Off-Axis Digital Holographic Microscopy

Jan - Jun 2018 | Umeå, Sweden

IAESTE Internship at University of Zagreb

Transport at Low Temperatures and High Magnetic Fields

Sep - Dec 2019 | Zagreb, Croatia

Awards

IRTG Community Prize (Best Fail)

Award for openly presenting a well-documented negative result and the lessons learned

2024 | Konstanz, Germany

Best Poster Community Prize

WE-Heraeus Seminar 834 "Superconducting Quantum Circuits Meet Quantum Materials"

2025 | Bad Honnef, Germany

Skills

Sample Preparation:

- Electron beam lithography (EBL)
- Shadow evaporation (Al/AlOx/Al)
- Reactive ion etching (RIE)
- Device contacting (silver paint)

Cryogenic & Microwave Instrumentation

- Dilution refrigerator operation
- MCBJ mechanics / alignment
- Low-noise DC wiring & thermalization
- RF/microwave wiring & VNA (GHz)
- Automation & control software (p5control-bluefors)

Calibration & Systematics

- DC chain calibration (gain/offset/series R)
- Noise floor characterization (PSD)
- Reproducible logging and analysis pipelines (p5control-bluefors-evaluation)

Modeling & Parameter Extraction

- Josephson dynamics: RSJ/RCSJ
- Microwave response: Shapiro steps and photon-assisted tunneling (PAT)
- MAR-based transport modeling
- SciPy / NumPy / JAX implementations (superconductivity)

Hobbies

Music
French Horn, Choir

Nature
Cycling, Trekking, Travel

Canoe Polo
Seelöwen Konstanz e.V.

Supervision

Patrick Raif

- Project Report: Fabrication of a Superconducting Single-Electron Transistor
- Master thesis: Electronic Transport through Superconducting Atomic Contacts under Microwave Irradiation

Apr 2022 - Mar 2024 — Master Student

Florian Kayatz

Software Development of a Measurement Platform (mawenzy/p5control)

Jan 2023 - Apr 2023 — Research Assistant

David Capalbo

Project Report: Simulations on Photon-Assisted Multiple Andreev Reflections

Apr 2025 - Aug 2025 — Master Student

Conferences

Frontiers of Condensed Matter

Summer School

Okt 2022 — Les Houches, France

Tunable Superconducting SET: from Weak- to Strong Coupling

- Poster at "DPG Spring Meeting"
  Sep 2022 — Regensburg, Germany
- Poster at "SFB1432: Fluctuations and Non-Linearities"
  Jan 2023 — Irsee, Germany

Superconducting Atomic contacts under microwave irradiation

- Talk at "DPG Spring Meeting"
  Mar 2025 — Regensburg, Germany
- Poster at 834. WE-Heraeus-Seminar
  "Superconducting Quantum Circuits Meet Quantum Materials"
  May 2025 — Bad Honnef, Germany

Publications

- **O. Aschenbrenner**, P. Raif, J. C. Cuevas & E. Scheer, *Current biased superconducting atomic contacts under microwave irradiation* (in preparation).
- L. Schertel, **O. Irtenkauf**, C. M. Aegerter, G. Maret & G. J. Aubry, *Magnetic-field effects on one-dimensional Anderson localization of light*, Physical Review A 100, 043818 (2019).